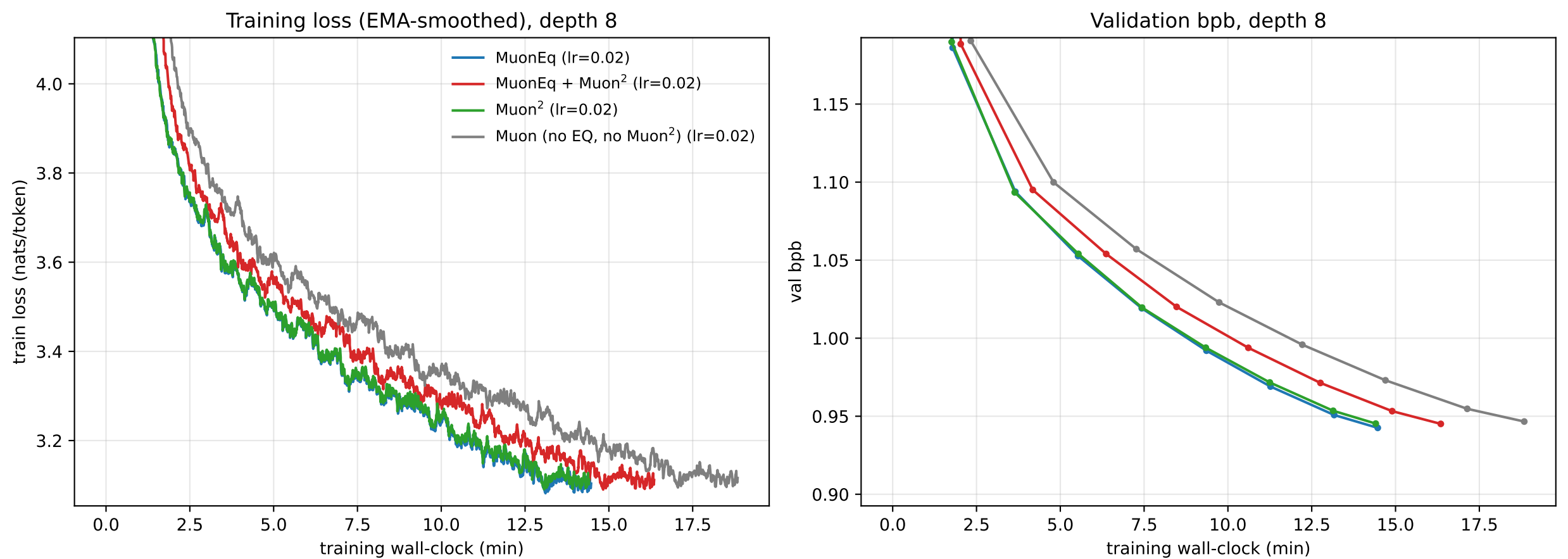

Training loss (EMA-smoothed), depth 8
MuonEq (lr=0.02)
MuonEq + Muon² (lr=0.02)
Muon² (lr=0.02)
Muon (no EQ, no Muon²) (lr=0.02)
train loss (nats/token)
4.0
3.8
3.6
3.4
3.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
training wall-clock (min)
Validation bpb, depth 8
val bpb
1.15
1.10
1.05
1.00
0.95
0.90
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
training wall-clock (min)